Tutorial 3

1.

(a) In a two-class classification problem, the training samples are :

Class +1 $\{(2, 1), (4, 2), (6, 5)\}$

Class –1 $\{(1, 7), (3, 4), (4, 8), (5, 6)\}$

(i) Identify the support vectors from the graph of the above training samples.
(ii) Find the optimal separating line (hyperplane).
(iii) How would the optimal line classify the unknown with feature (3,2) ?

(b) (i) If for a 3D feature space, the optimal separating hyperplane $h(\boldsymbol{x})$ is given by

$$h(\boldsymbol{x}) = x_1 + 2x_2 - x_3 + 4 = 0$$

where $\boldsymbol{x} = [x_1, x_2, x_3]^T$. Given a set of training examples for each class, explain how you can find the support vectors from each class.

(ii) How far is $[3, -1, 2]^T$ from the optimal separating hyperplane ?

(iii) How do you find the gap of the two supporting hyperplane?

2.

The pattern classes, ω_1 and ω_2, with training samples

ω_1 : $\{(-1, 0)^T, (0, -1)^T, (1, 0)^T, (0, 2)^T\}$

ω_2 : $\{(-2, 0)^T, (0, -2)^T, (2, 0)^T, (0, 6)^T\}$

are non-linearly separable. It is possible to apply a kernel transformation from the original input space into an abstract feature space that will separate the two classes linearly. For example, try the following transformations :

$\phi_1(\boldsymbol{x}) = e^{-\|\boldsymbol{x} - \boldsymbol{t}_1\|^2}$ and $\phi_2(\boldsymbol{x}) = e^{-\|\boldsymbol{x} - \boldsymbol{t}_2\|^2}$

Set $\boldsymbol{t}_1 = (0, 0)^T$ and $\boldsymbol{t}_2 = (0, 2)^T$.

where are the support vectors, by observation, in the transformed abstract feature space ? Then calculate the optimal line in the abstract (transformed) feature space.

[Optional] Do you think the method in part (b) will still work if we had set $\boldsymbol{t}_1 = (0, 0.2)^T$ and $\boldsymbol{t}_2 = (0, 1.8)^T$? How should $\boldsymbol{t}_1$ and $\boldsymbol{t}_2$ be chosen ?

3. The following plane in 2D was found from training set,

$x + y - 3 = 0$,

for a 2-class data set, and

$$\begin{cases} class+1, & h(\boldsymbol{s}) > 0 \\ class-1, & h(\boldsymbol{s}) < 0 \end{cases}$$

(a) Giving support vectors for
class+1 = (0,5), (1,4)
class-1 = (0,1)
Derive the hyper-plane.
(b) Take an input of $\boldsymbol{s}$ = (1, 1), determine its class.
(c) Repeat (b) for data input (5,9) & (-1,-2)

(d) For the same plane, if support vectors are
class+1 = (0,6), (2,4)
class-1 = (-1,1)
What is the hyper plane?

4. Find the Euclidean distance of an input data $\boldsymbol{s}$ = (10, 20, 30) to the hyper-plane

$h(\boldsymbol{s}) = -5x + 10y + 112z + 57 = 0$

(a) Derive the distance equation
(b) Calculate the distance
(c) How to find the separation of two supporting hyper-planes?

Note: it is normal that if the distance is a negative number.

5. Why the template matching cannot be used for face recognition?